

Automatic take-off/landing strategies and control for a multi-kite Airborne Wind Energy system

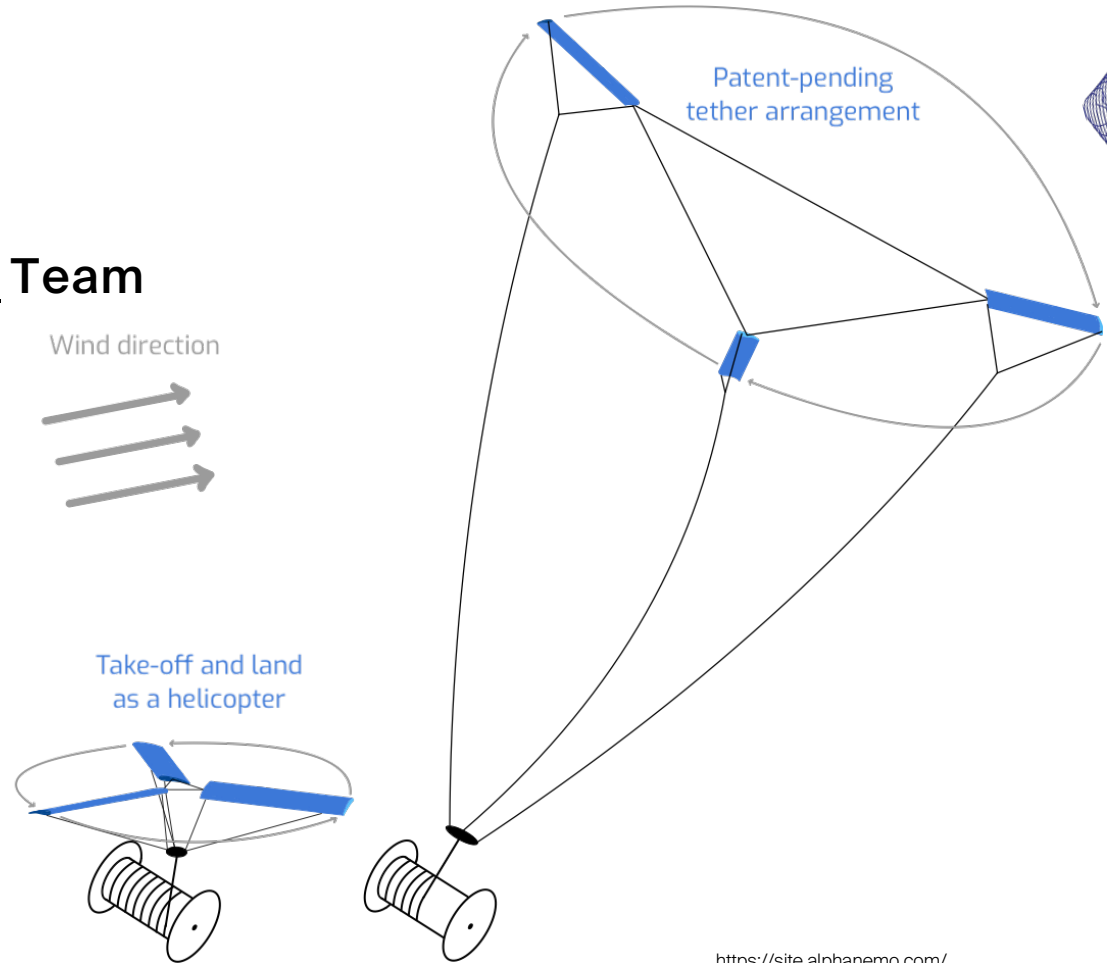
Supervision: J. Waibel, Prof. C. Jones



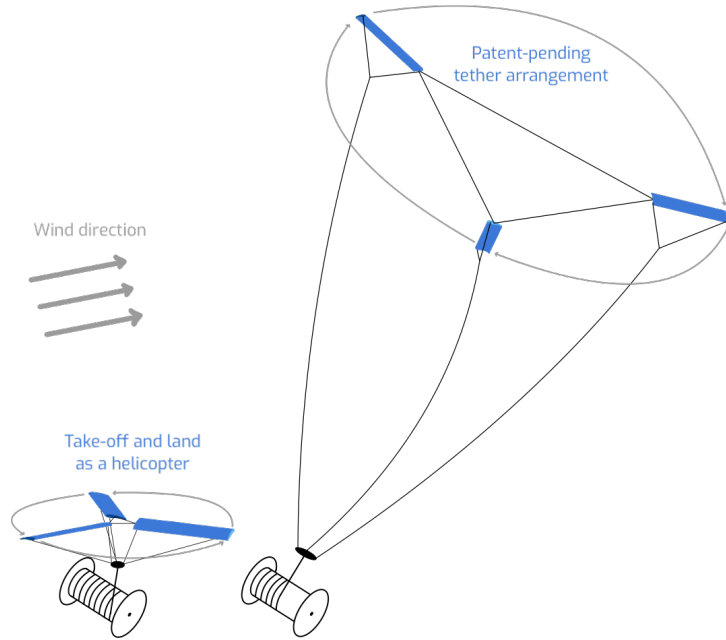
<https://site.alphanemo.com/>

Context and introduction

- Continuation of the work of the AlphAnemo Team
- Develop MultiKite Energy Systems
- Industrial constraints



Context and introduction



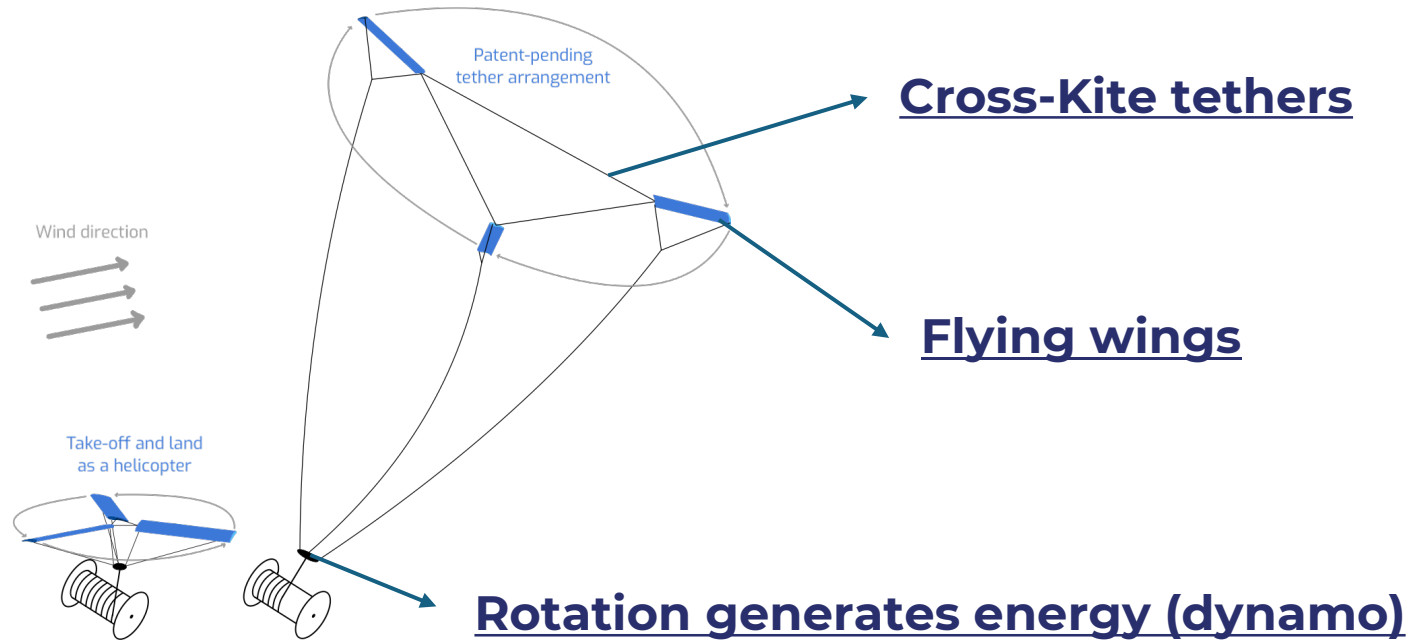
<https://site.alphanemo.com/>



https://pngtree.com/freepng/png-wind-turbines_14597478.html

- Large structure
- Large cost
- Large footprint (visual)
- Height limits force placement in highly visible areas

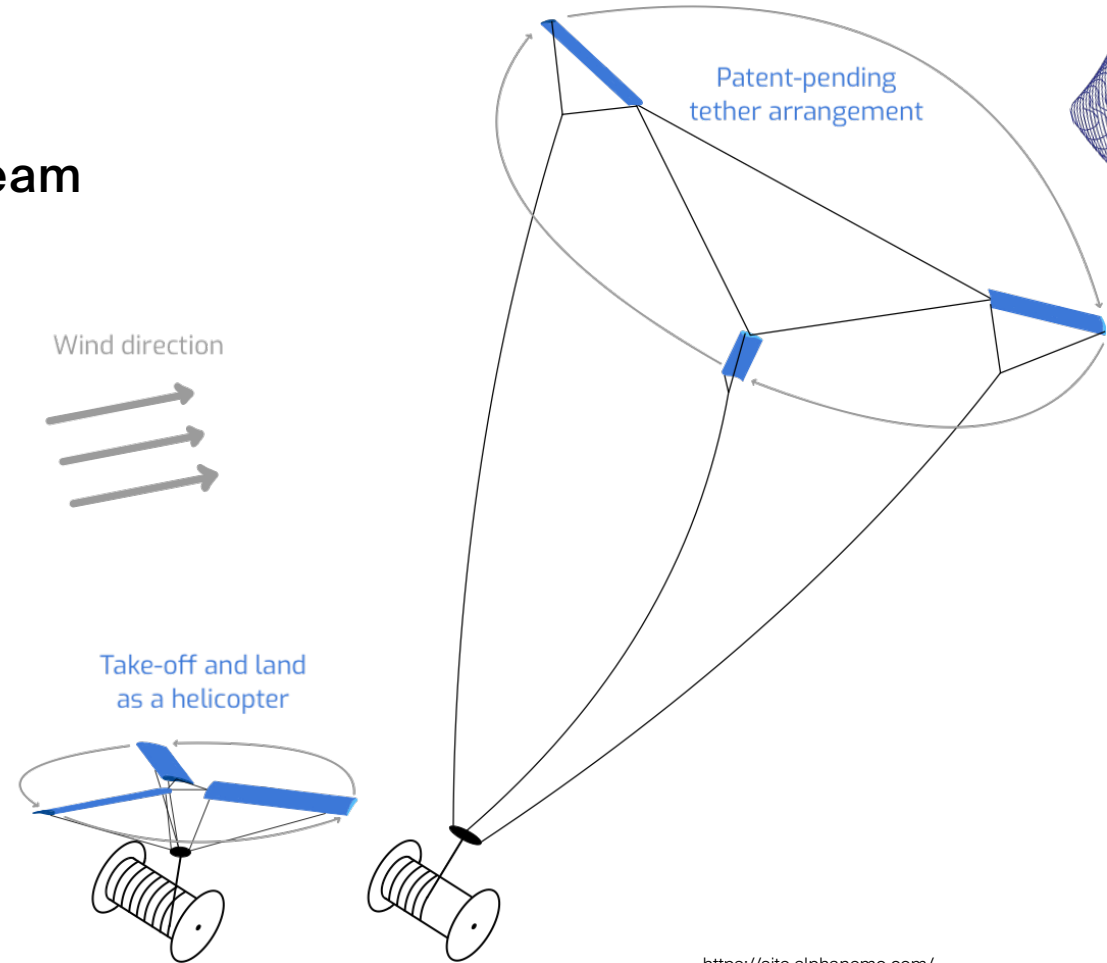
Context and introduction



<https://site.alphanemo.com/>

Context and introduction

- Continuation of the work of the AlphAnemo Team
- Develop MultiKite Energy Systems
- Industrial constraints
- Work in pseudo-parallel with another student



<https://site.alphanemo.com/>

Context and introduction

- Continuation of the work of the AlphAnemo Team



<https://www.linkedin.com/company/alphanemo/post>

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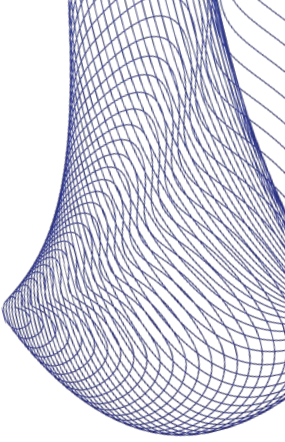
"chaotic take-off and landing"

Summary:

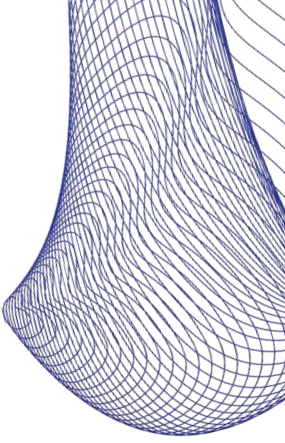
- Context and Introduction
- Goals
- First Major Version
 - Construction
 - Integration
 - Testings and results
 - Limitations and shortcomings
- Future work and conclusion

Goals

1. Familiarize with the existing desktop prototype and its behavior. (Videos)
2. Literature review on rotating take-off/landing (TOL) in AWE •
3. Generate options and assessment criteria
4. Simulate most promising concepts
 - a. Upgrade aerodynamic model for high angle-of-attack (spline model for coefficients)
 - b. Identify the coefficient curves for the small wings in the drone wind tunnel
 - c. Implement controllers/state machine/automation in simulation
5. Give inputs to hardware design of the new prototype
6. Test TOL experimentally



Goals

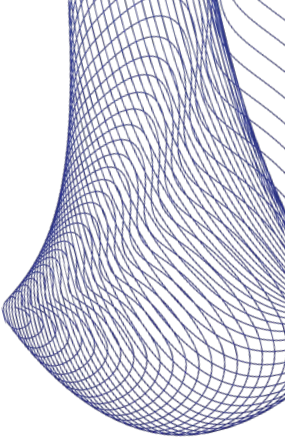


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Understand wing configuration

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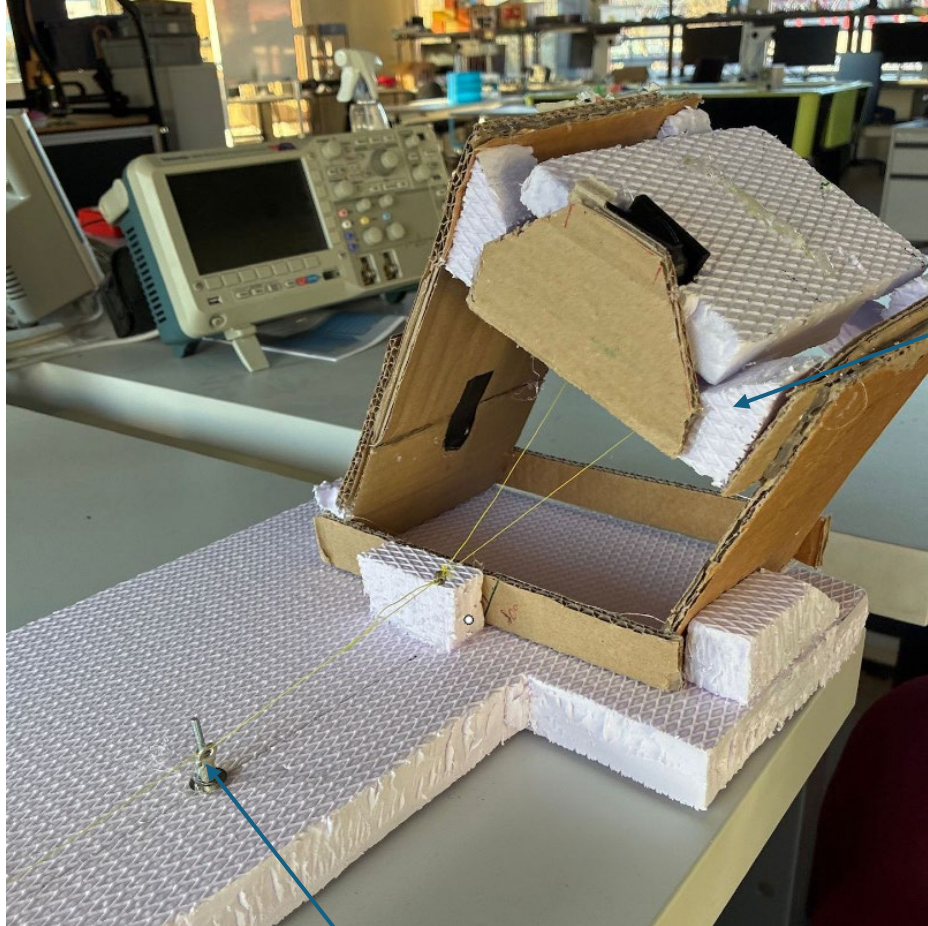


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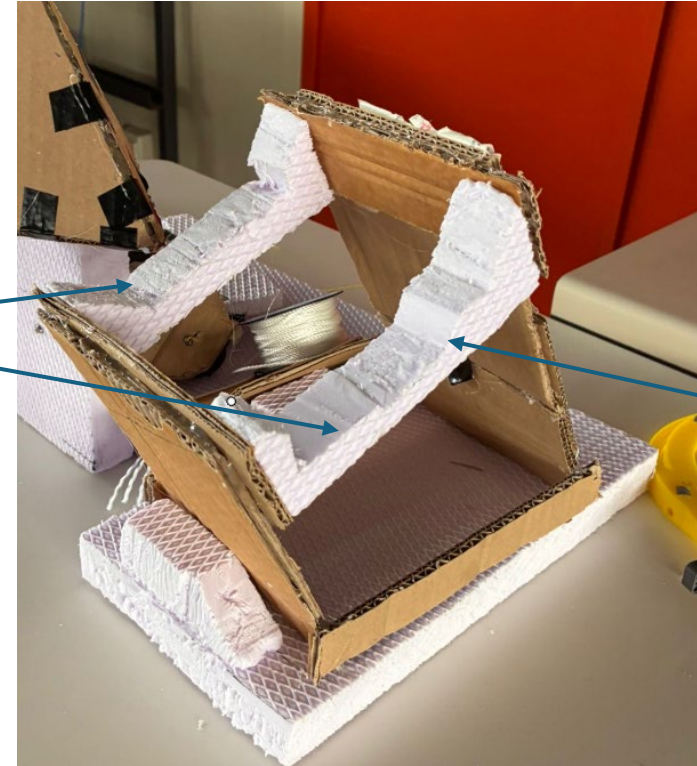
→ Version 0 created

Version 0

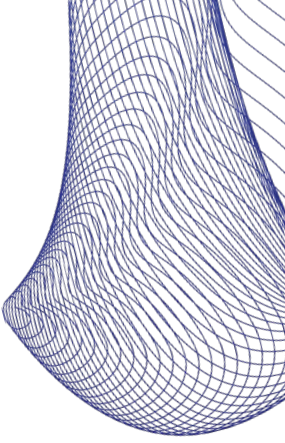


Tether attachment point

funnels

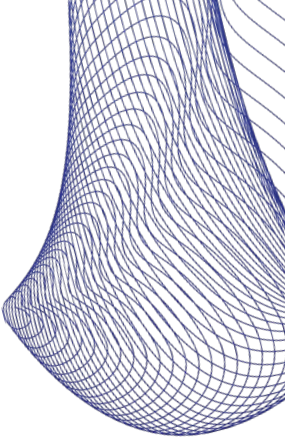


30° AOA



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Uneven surface

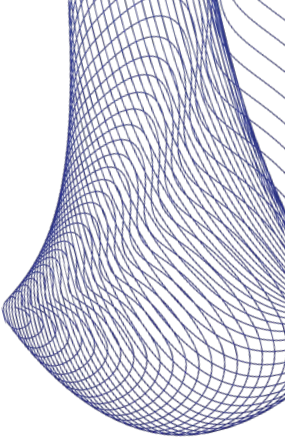
Tape + hanging masses

Goals

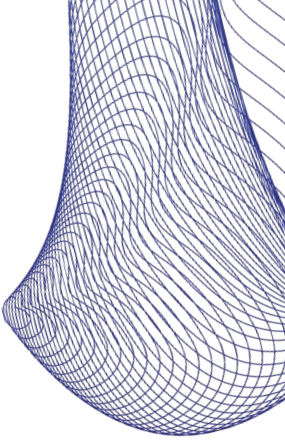
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standby

high uncertainty



Goals



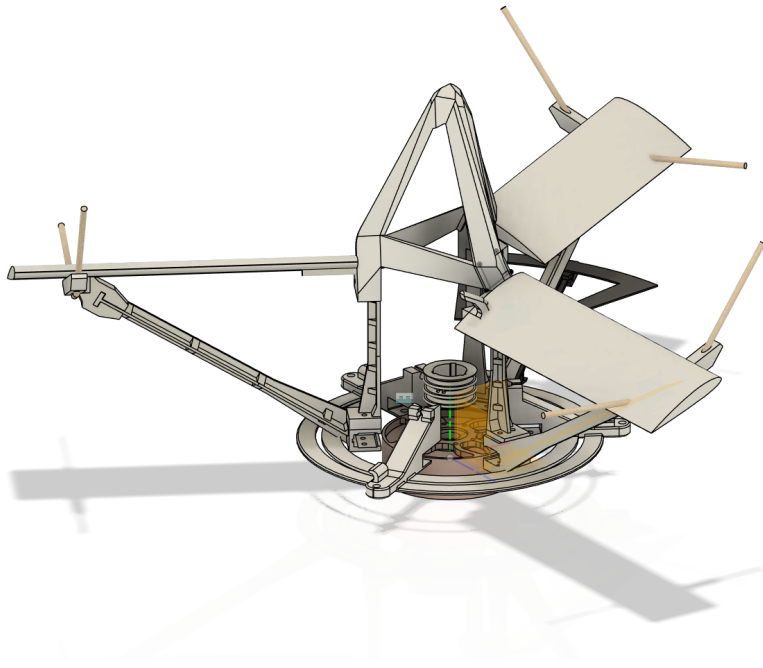
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Direct involvement

Current phase (version 1)

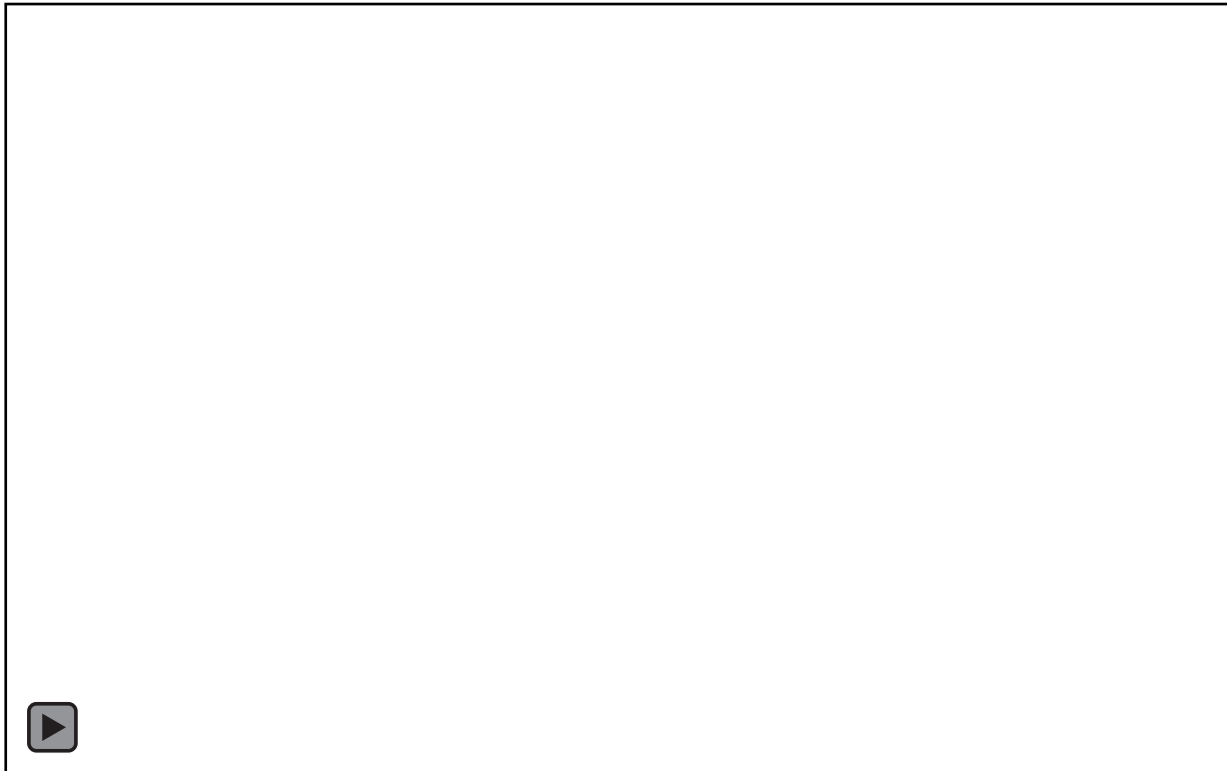
Version 1: Construction and integration

> Based on Version 0



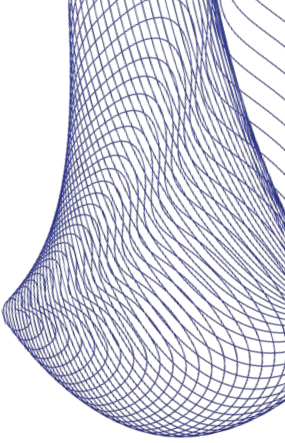
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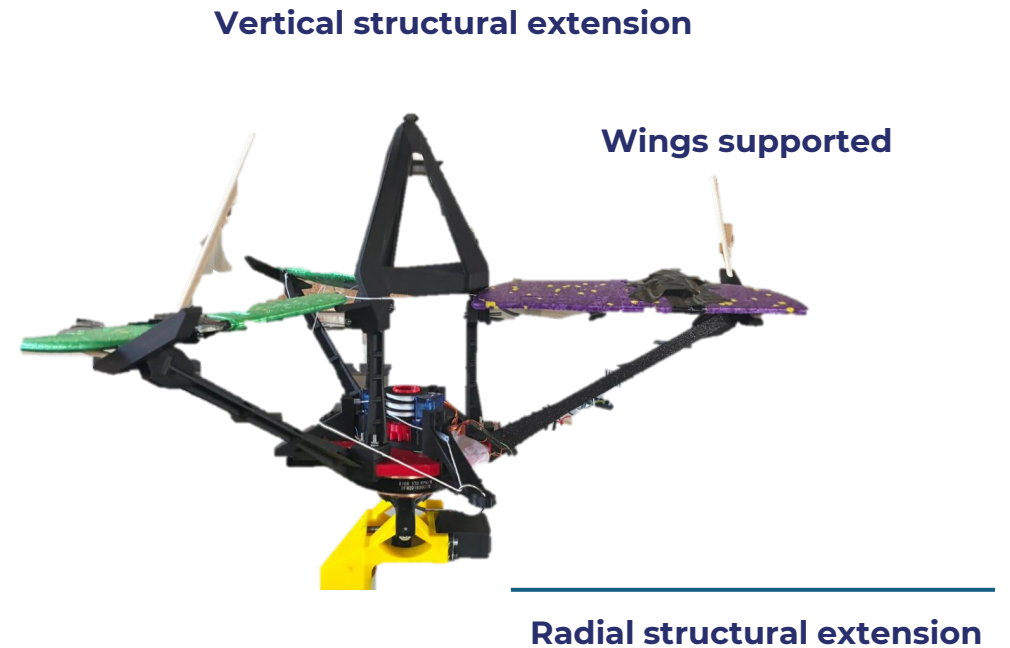


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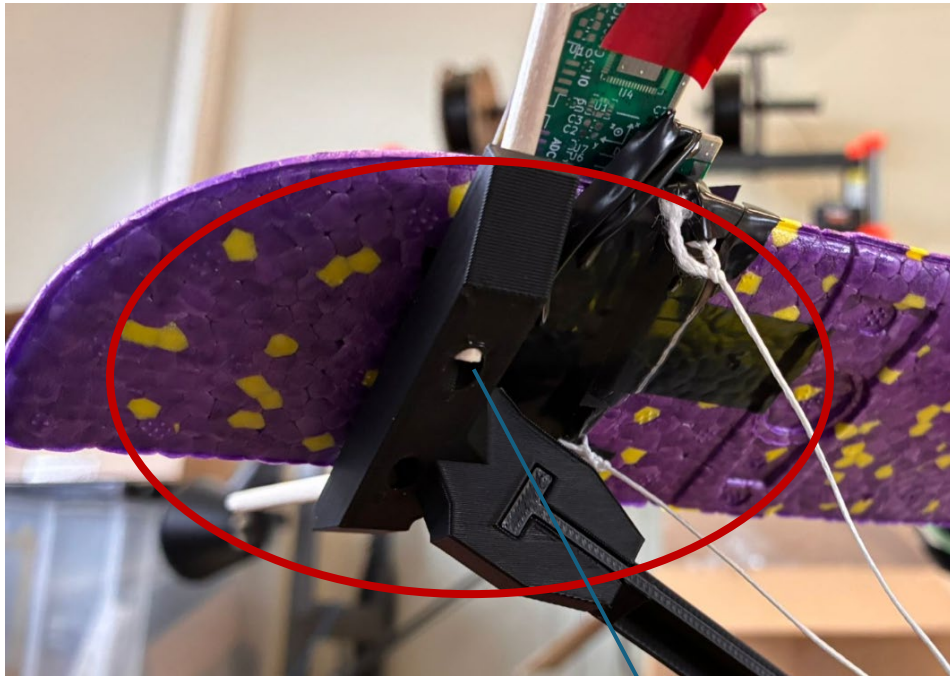


Old desktop prototype

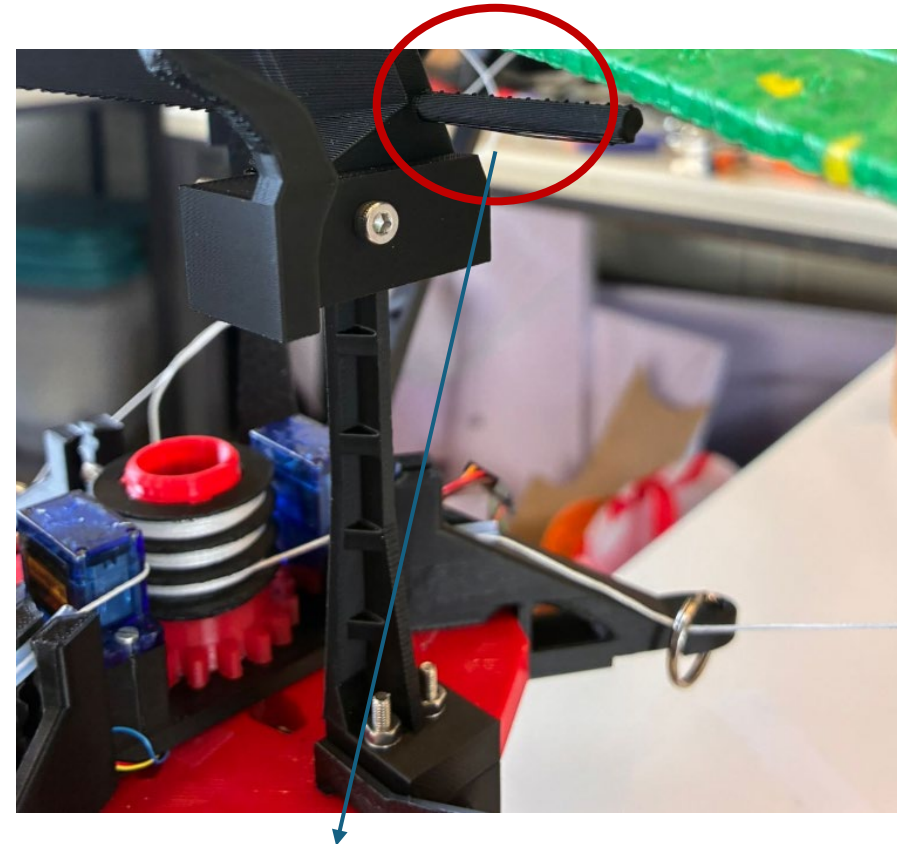


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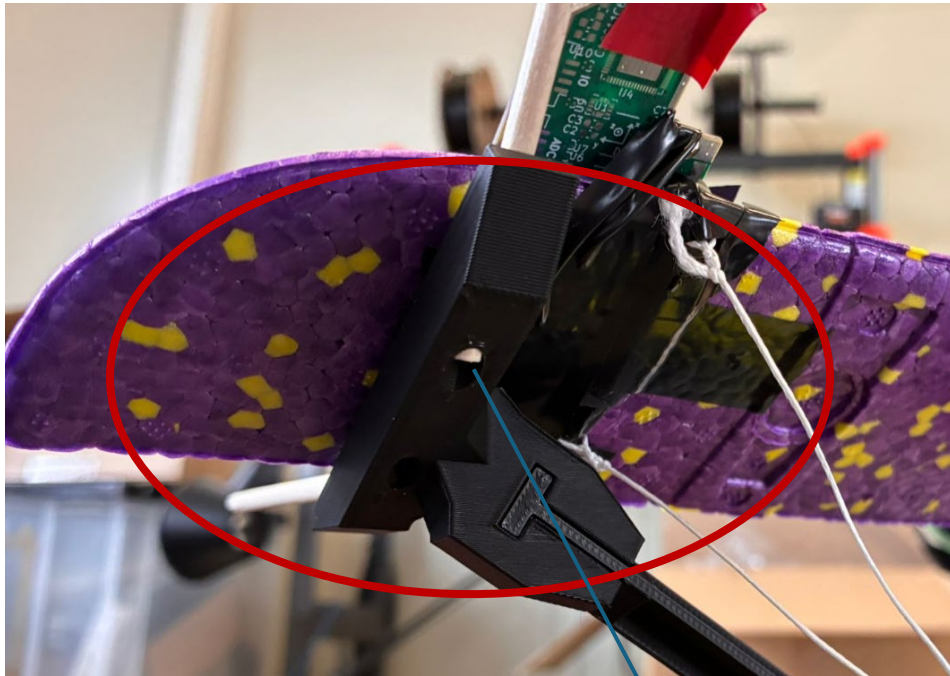
funnel



Contact point

Version 1: Construction and integration

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funnel

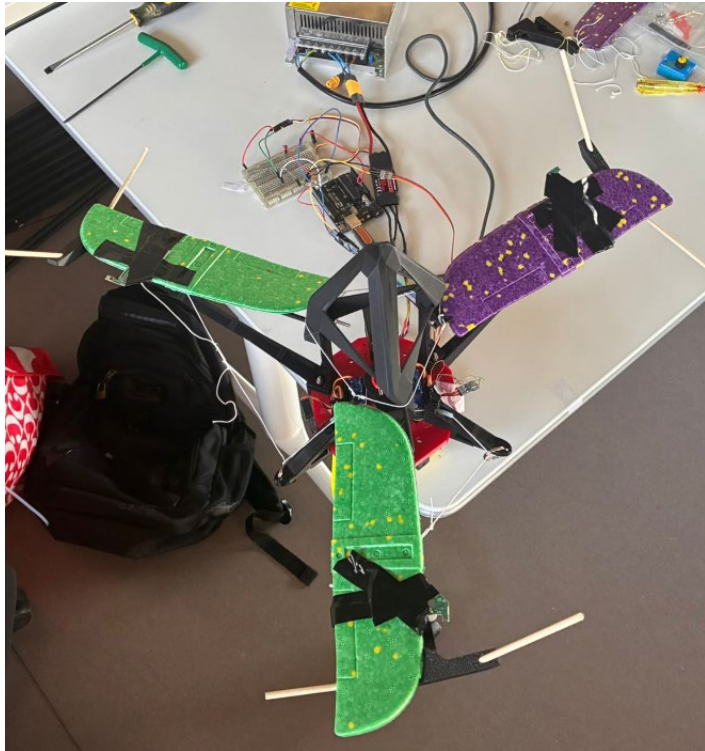
Half-funnel



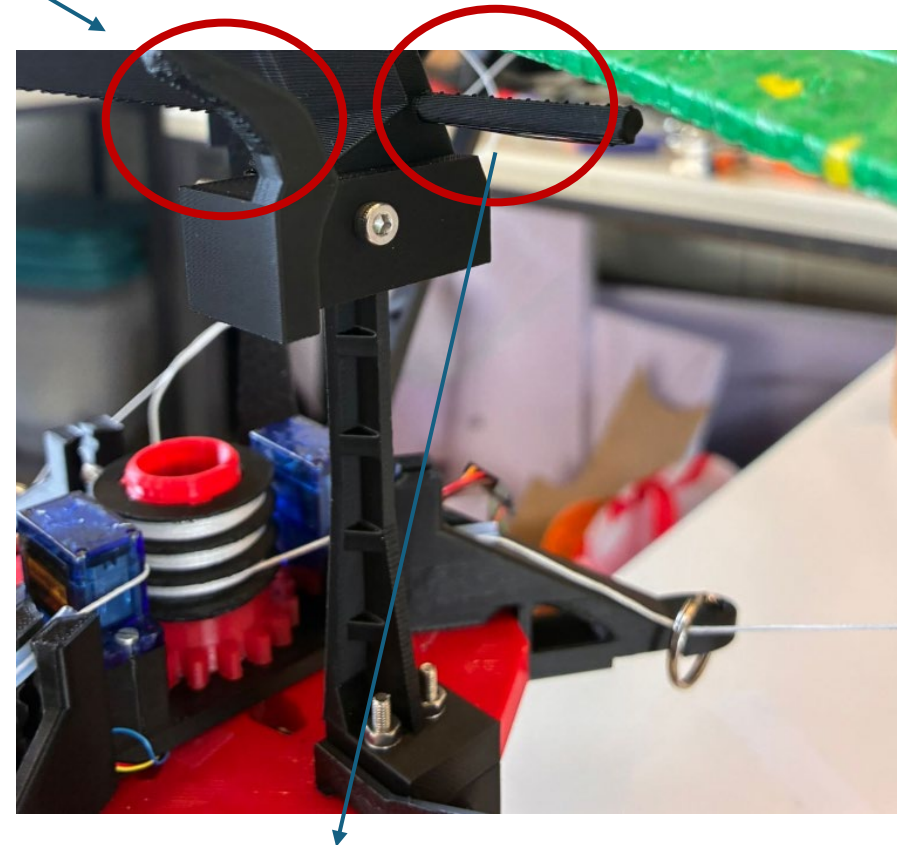
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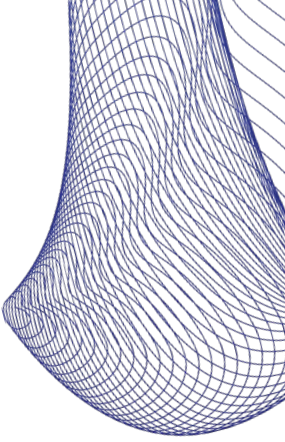
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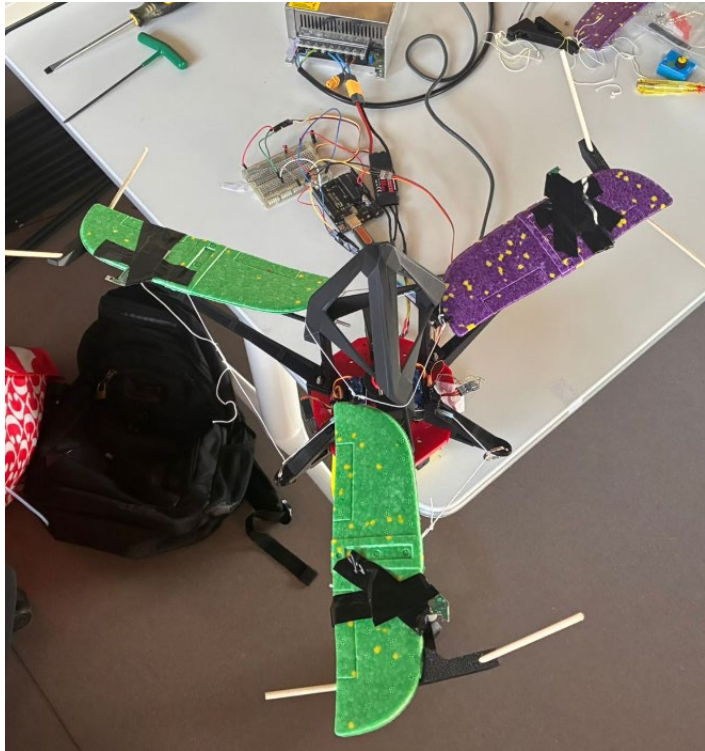


Contact point

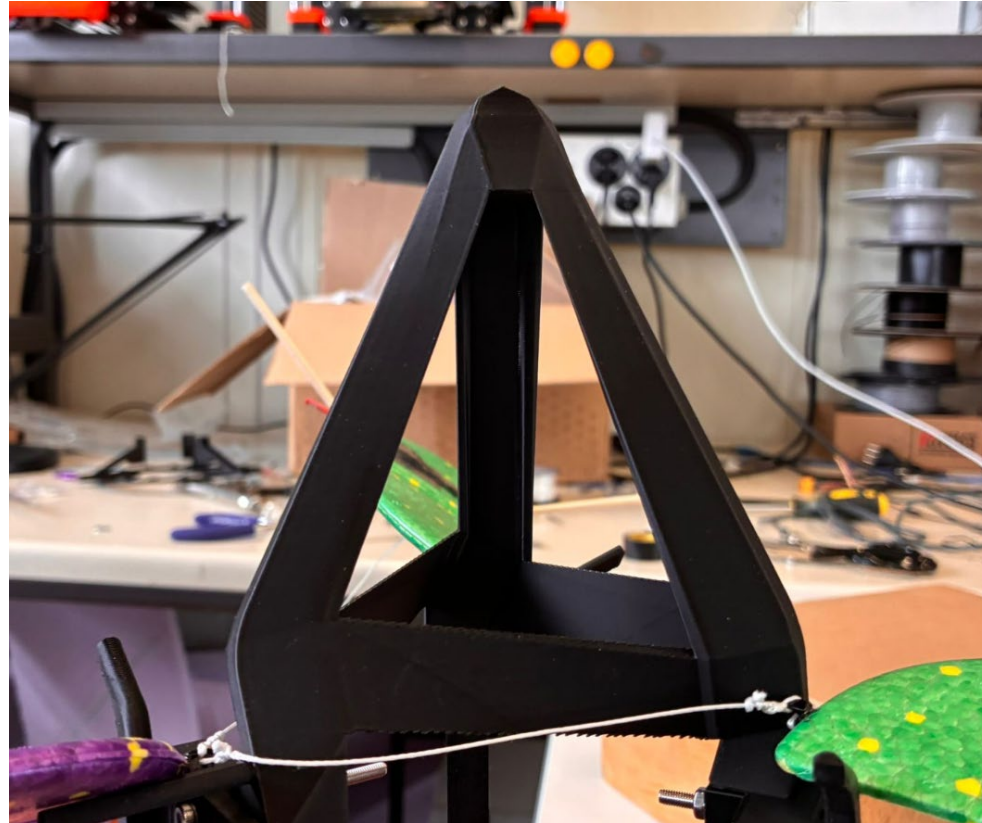


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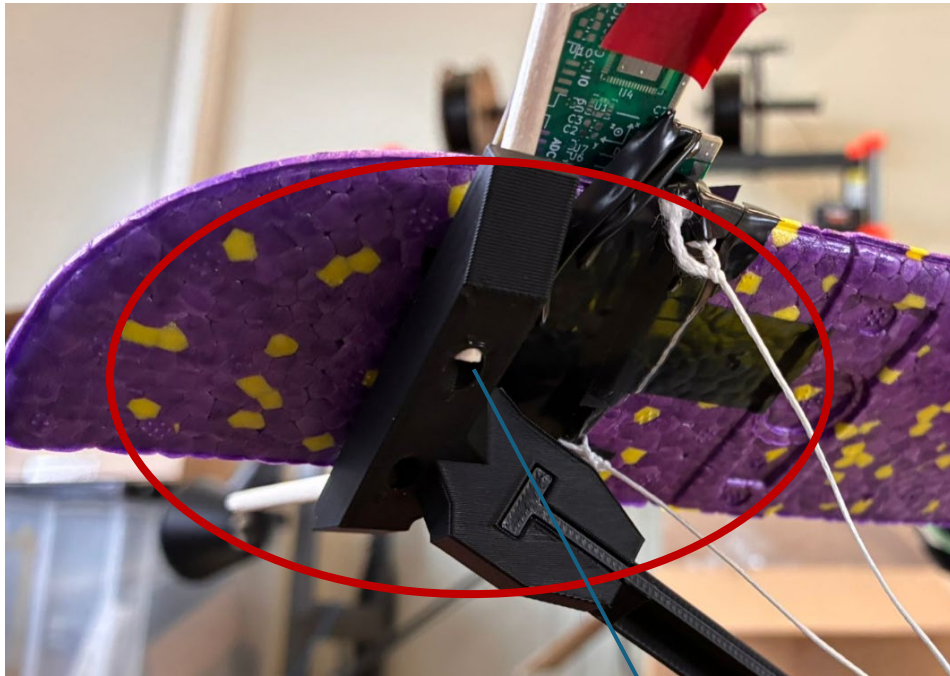


“pyramid”

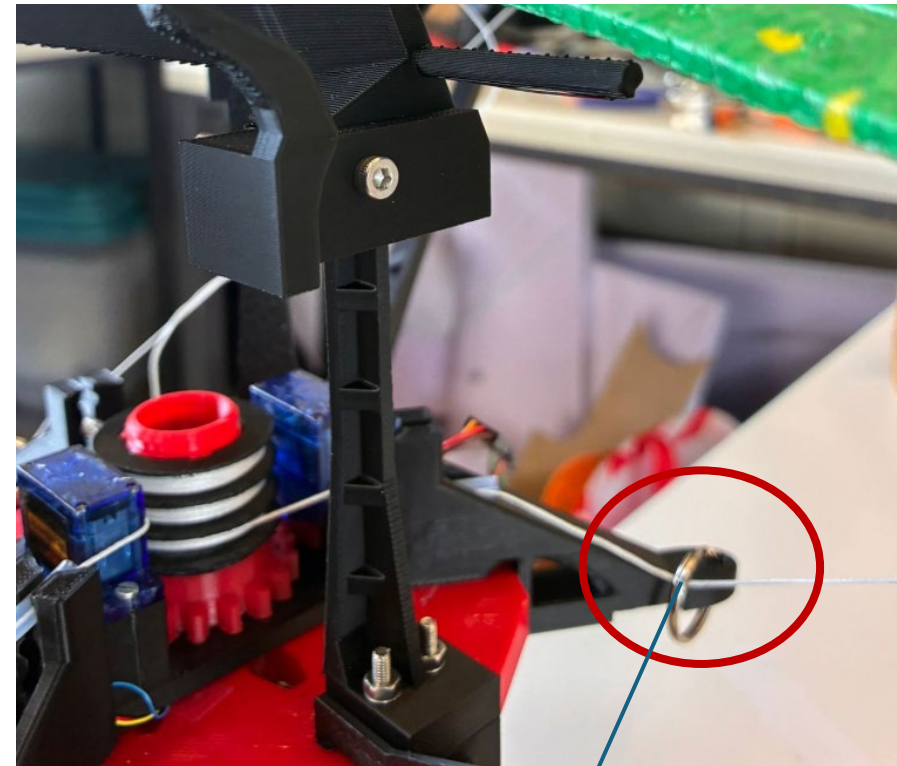


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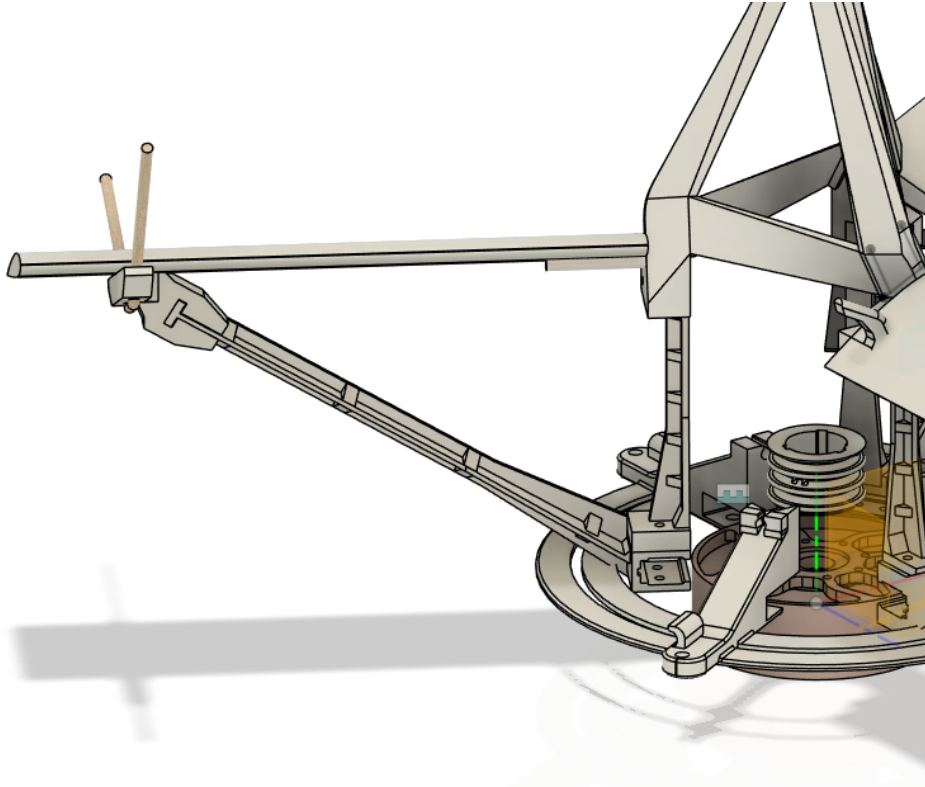
funnel



Attachment point

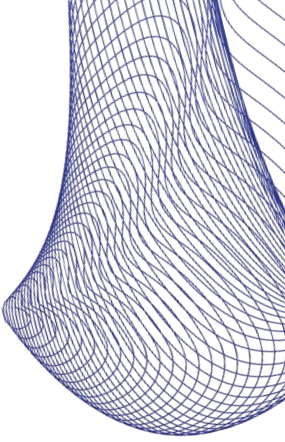
Version 1

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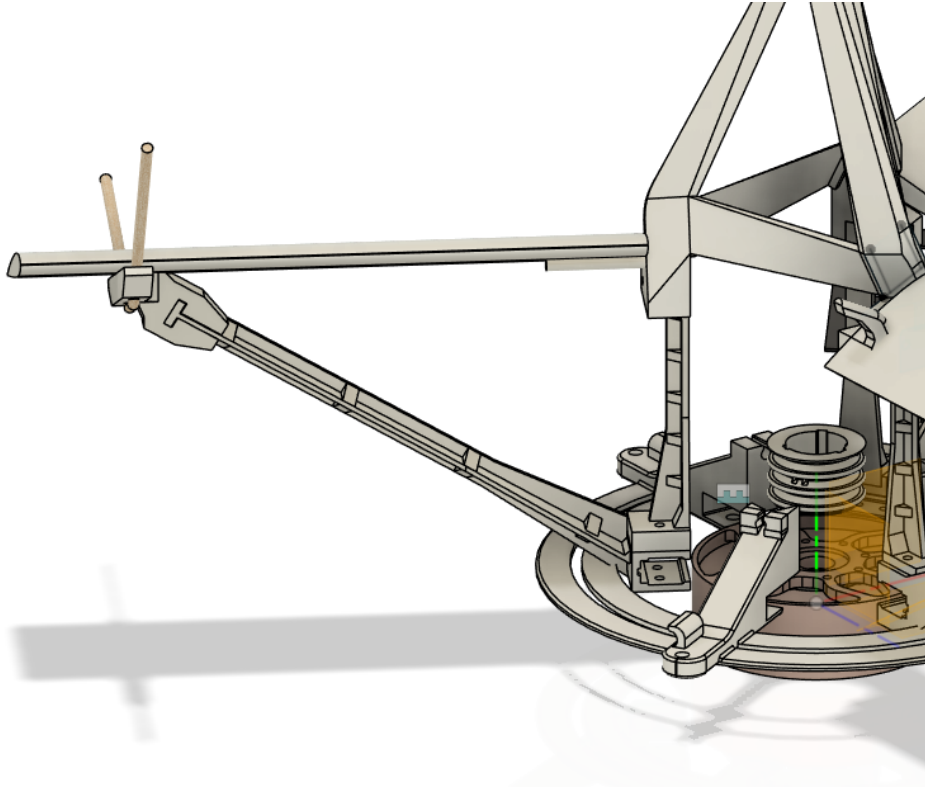
Arms:

- **Weight requirement**
- **Strength requirement**
- **Elongation requirement**



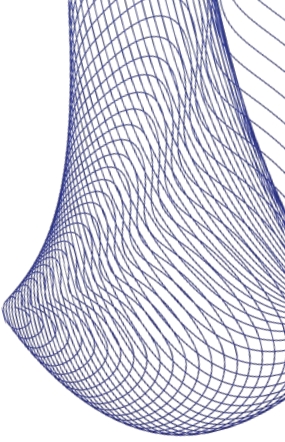
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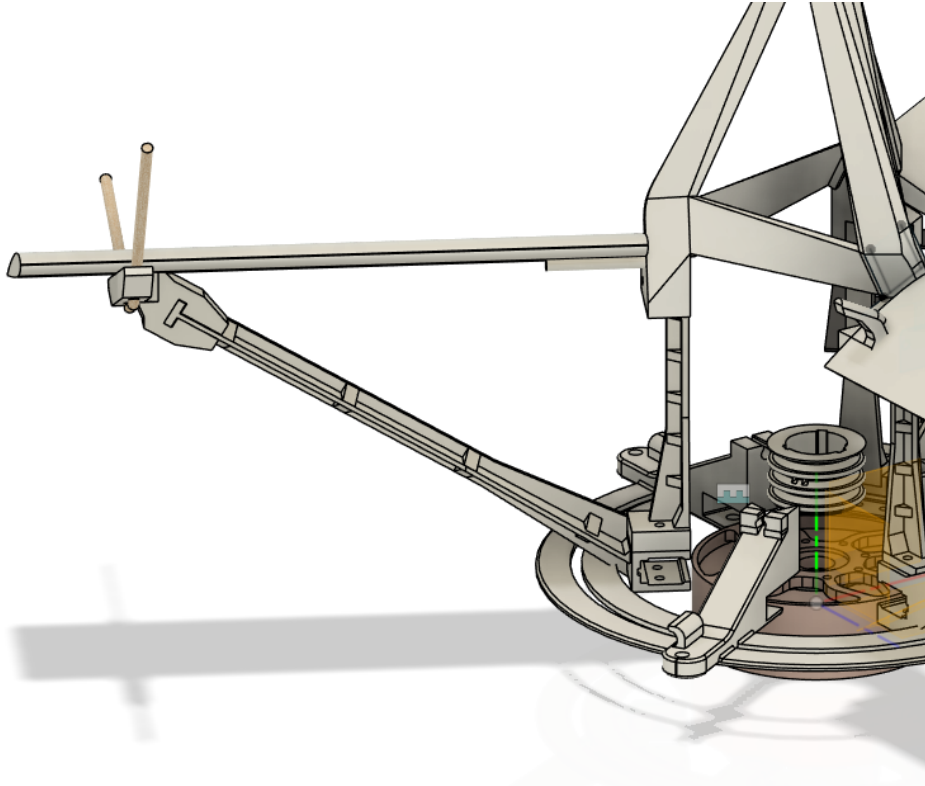
Arms:

- **Weight requirement**
- **Strength requirement**
- **Elongation requirement**
- **> T-shape**



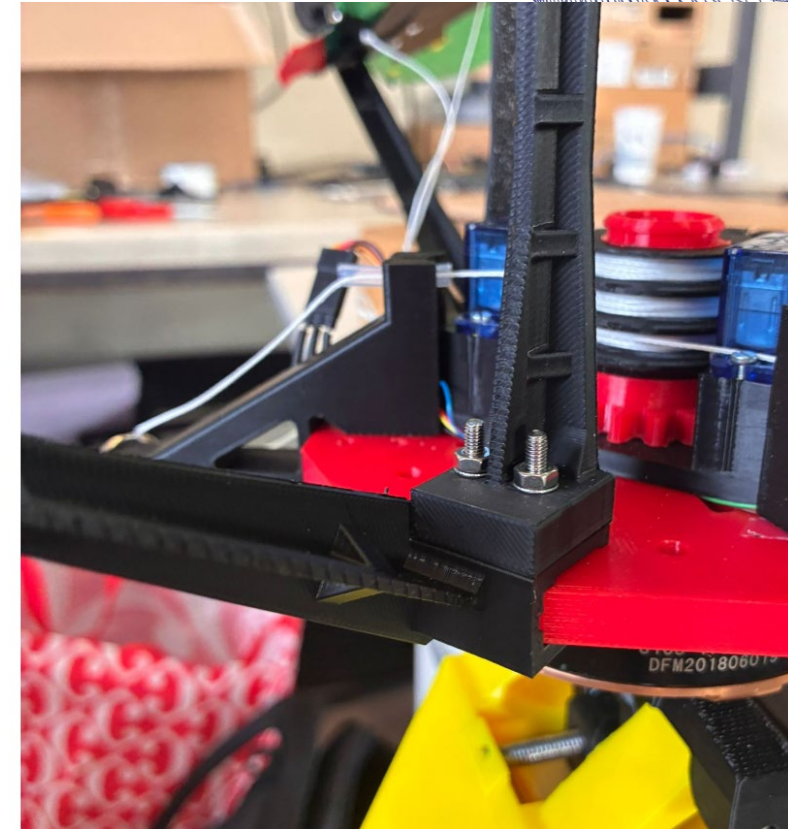
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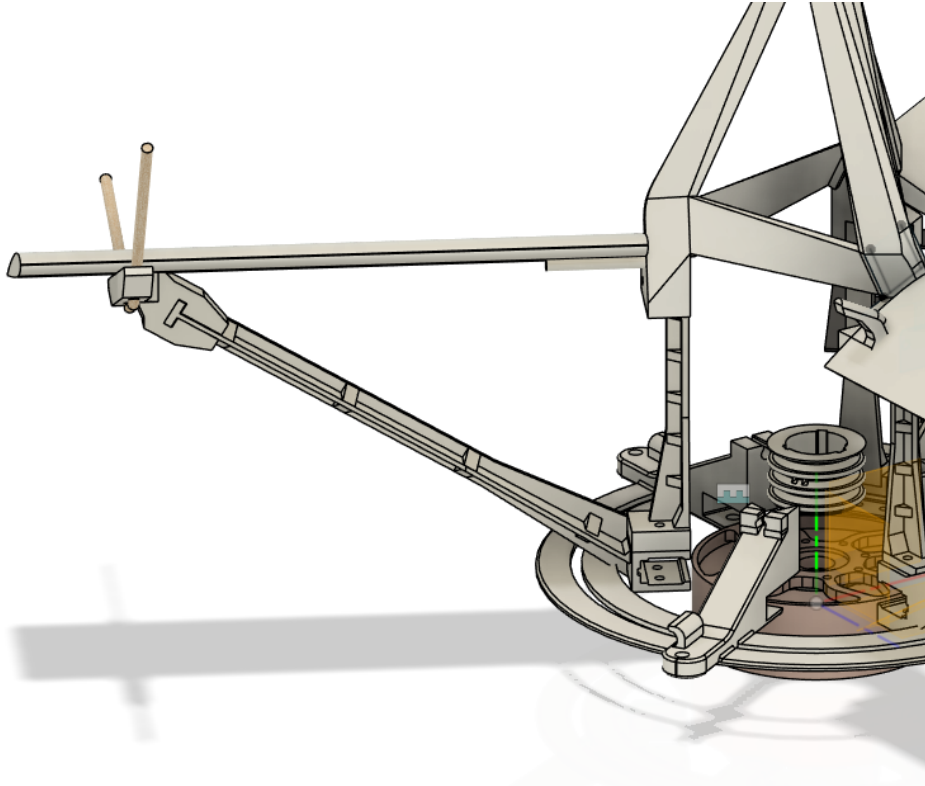
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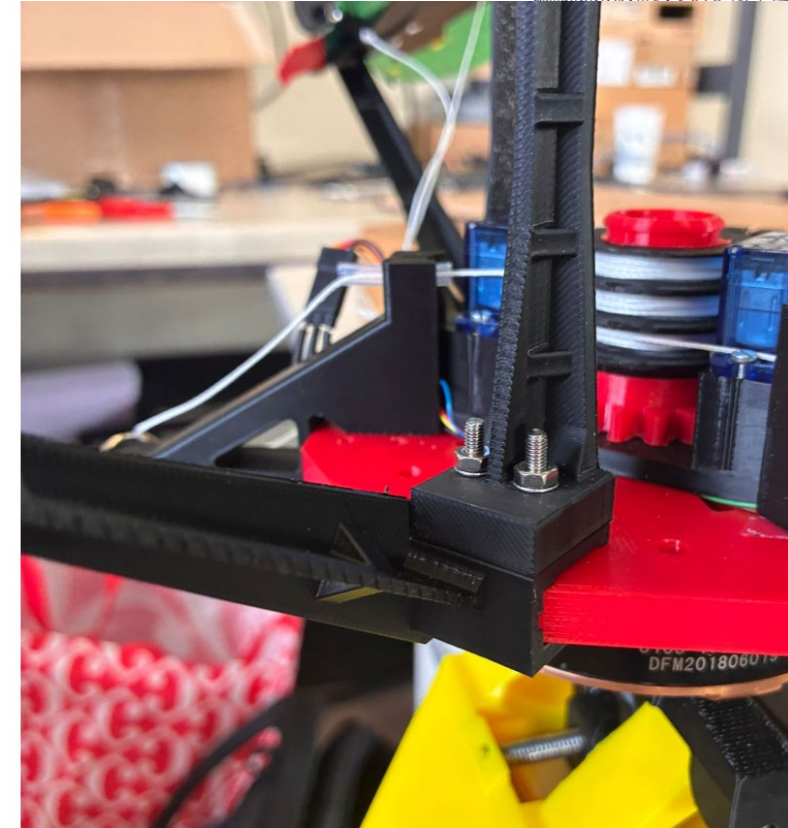
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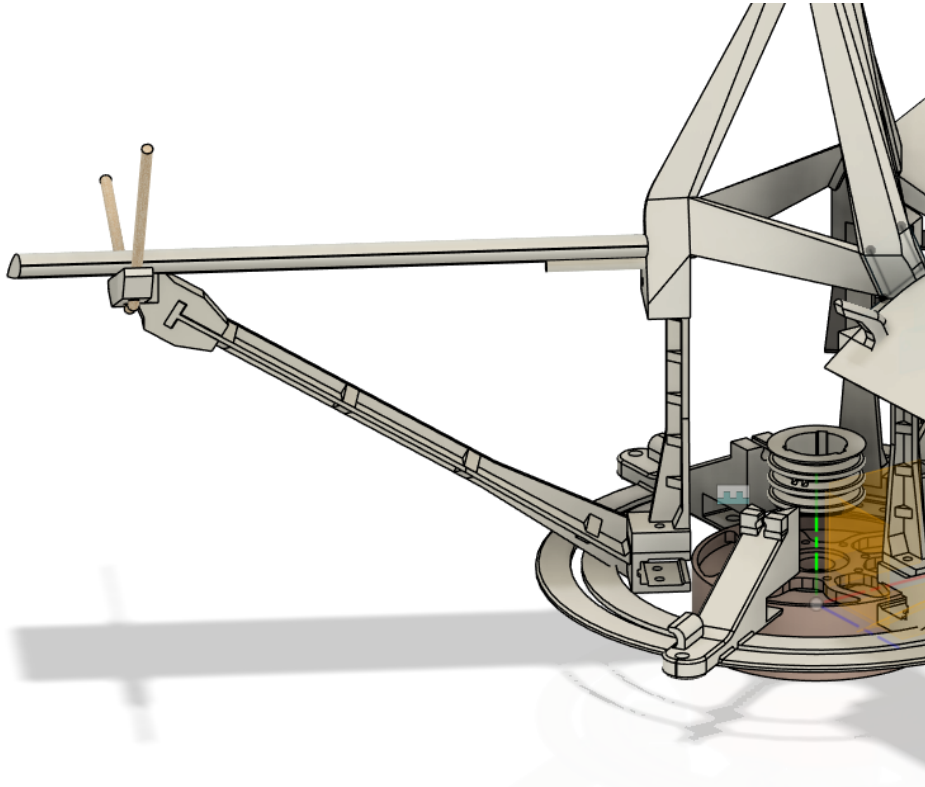
Arms:

- **Weight requirement**
- **Strength requirement**
- **Elongation requirement**
- **> T-shape**



Version 1

Based on Version 0



Arms:

- **Weight requirement**
- **Strength requirement**
- **Elongation requirement**

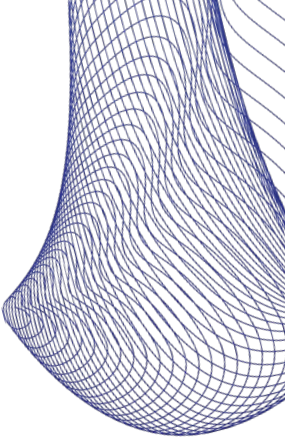


Version 1

Arms



- **Highly parametric**
- **Fast to print**
- **Lightweight**
- **Strong**
- **No failure under strong oscillations**

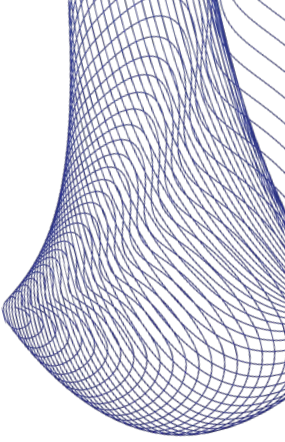


Version 1

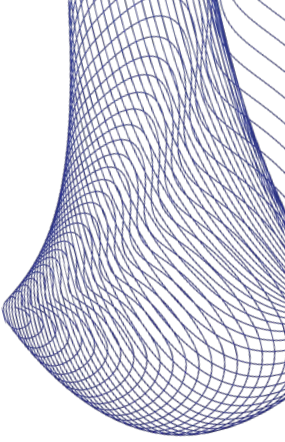
Arms



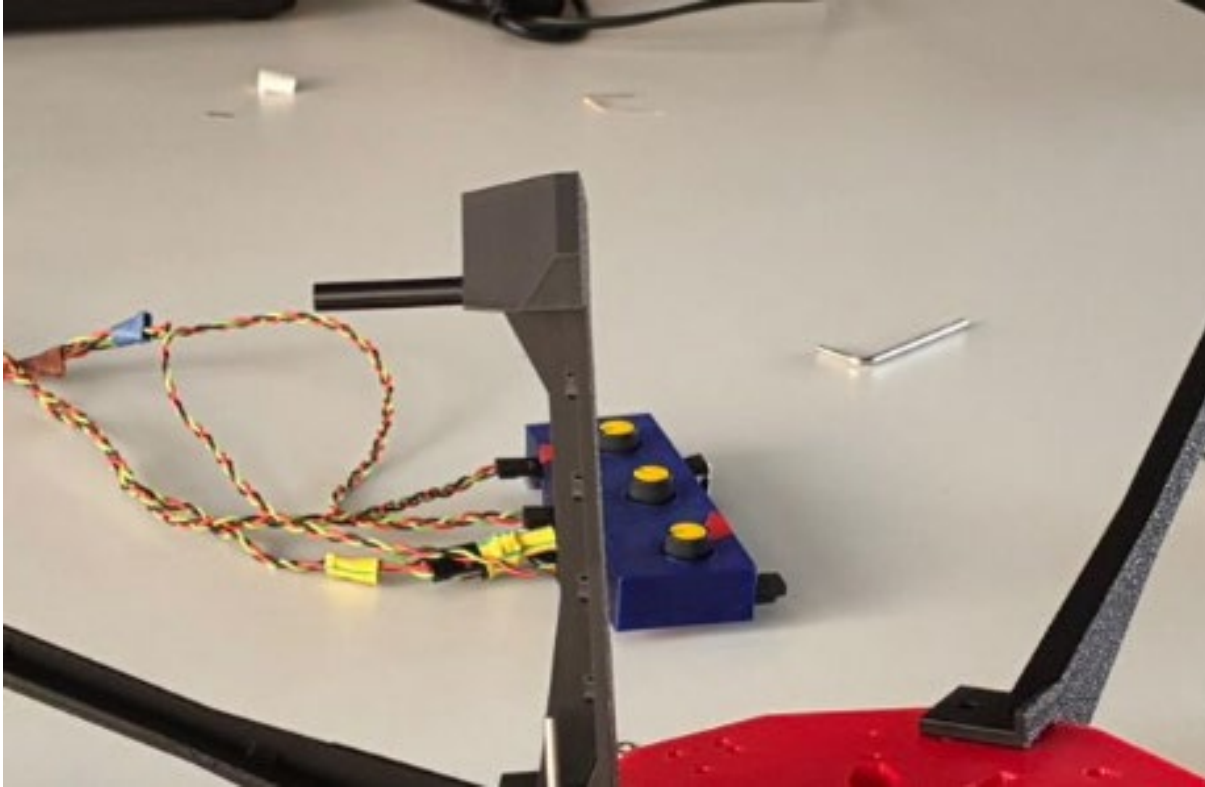
- **Highly parametric**
- **Fast to print**
- **Lightweight**
- **Strong**
- **No failure under strong oscillations**
- **Mounting the wrong arm does not have catastrophic results**



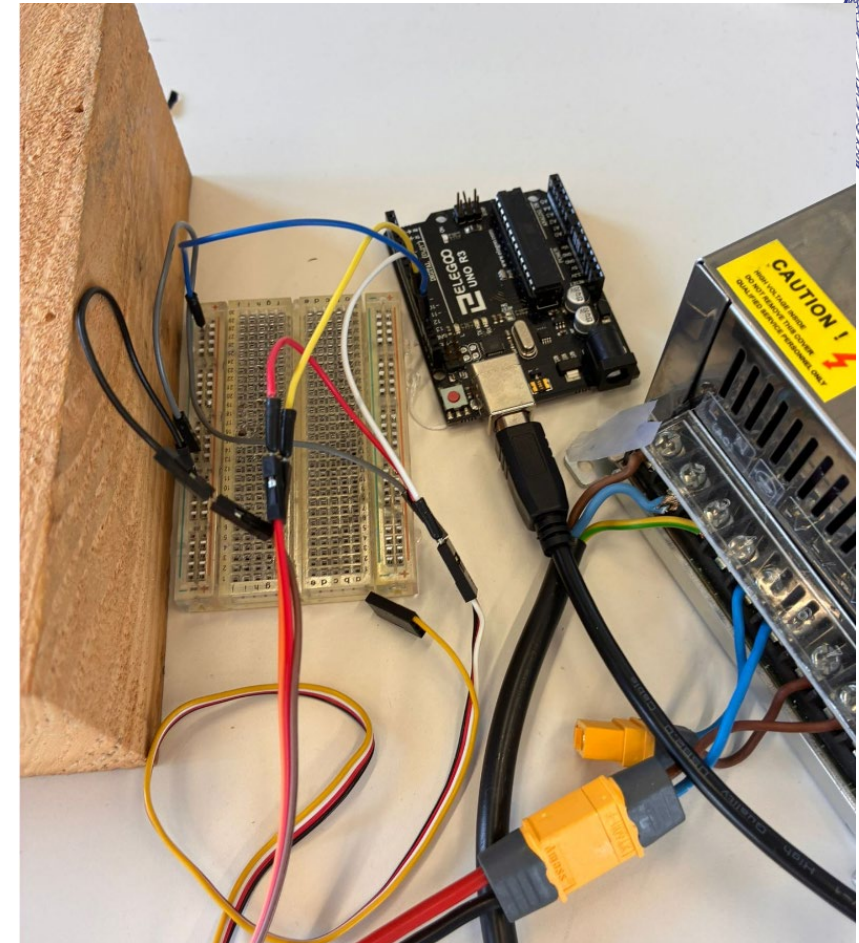
Version 1: construction



Integration: Electronics and software

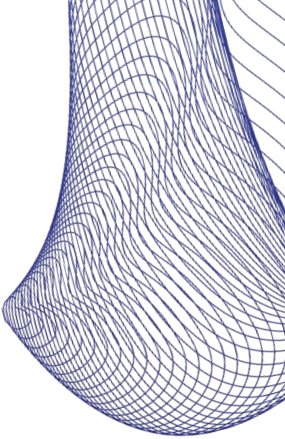


Before (manual)



Now

User interface



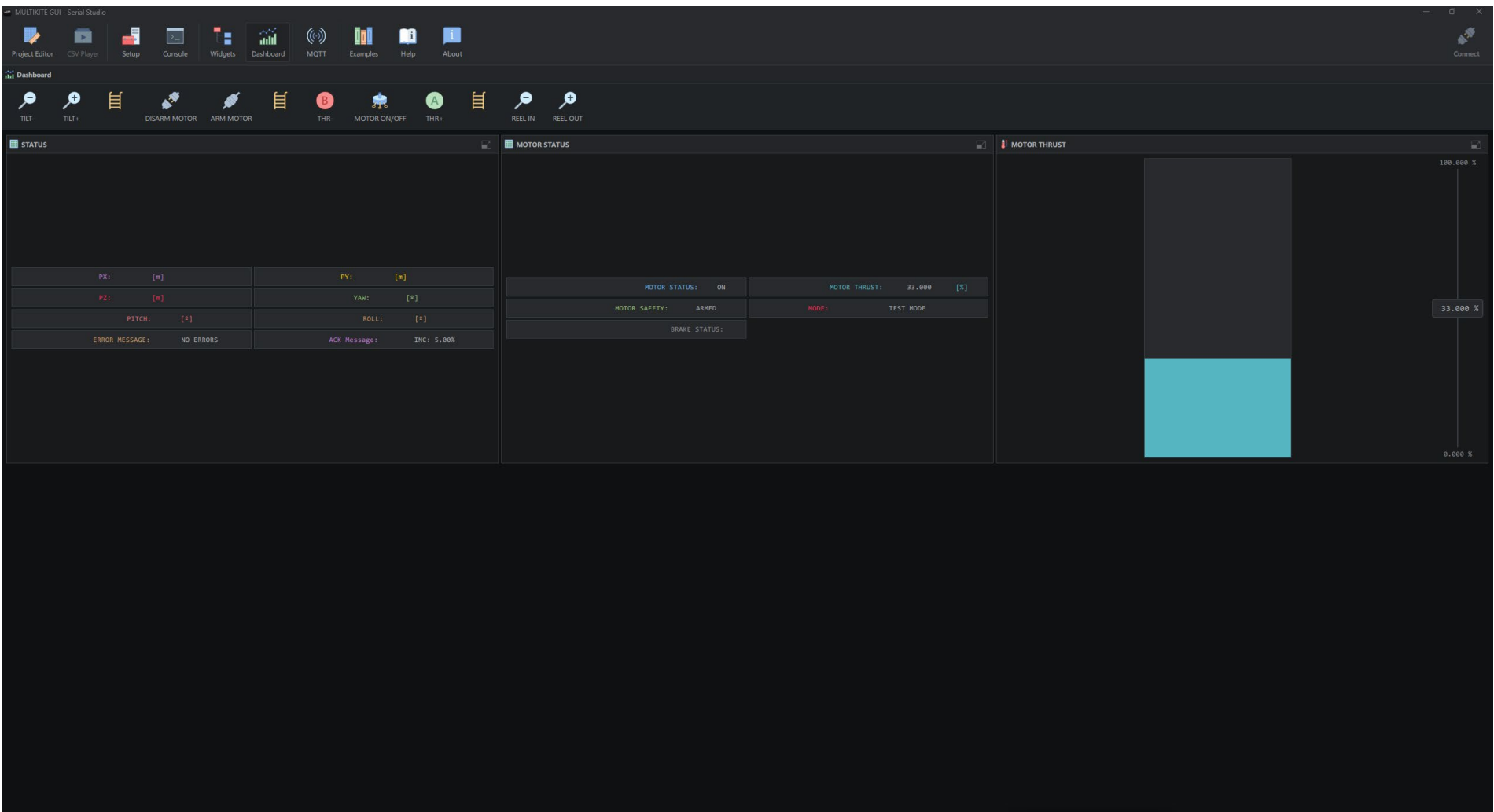
Quick action

Mimics real operation

Simplifies testing

Repeatable

Real time data



User interface



Repeatability

Ready for automatic procedures

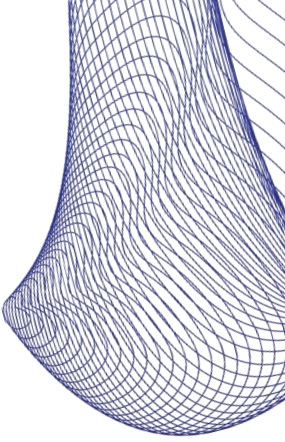
Modular

Log

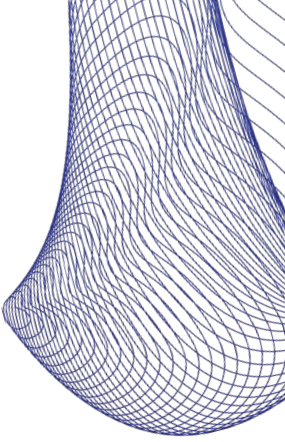
Safety (5m cable)

Testing and results

test



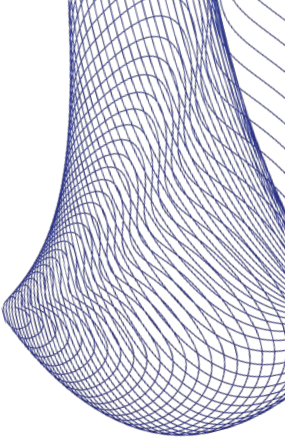
Testing



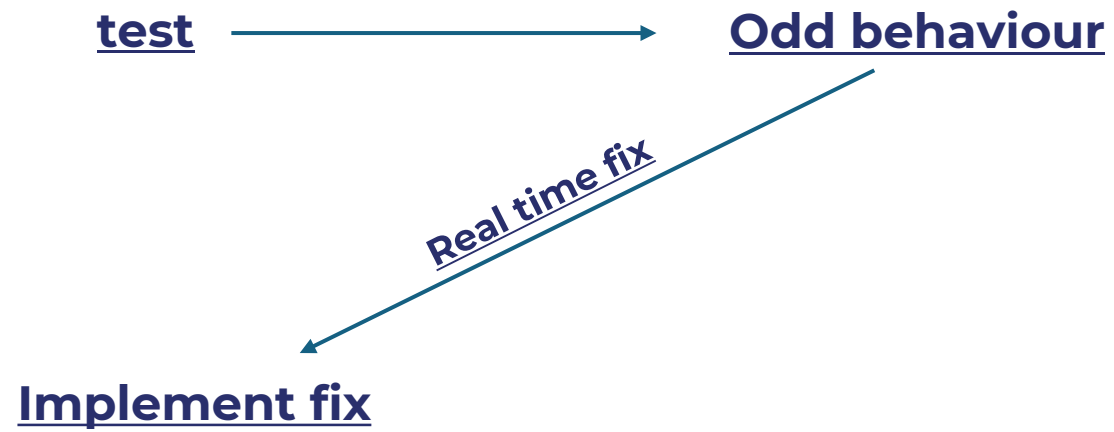
Issue example: one wing is pitched down

Testing

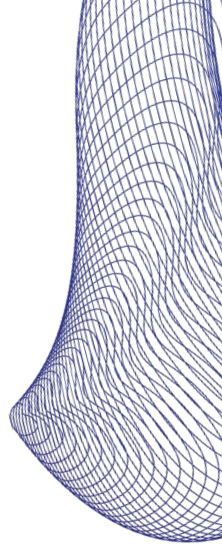
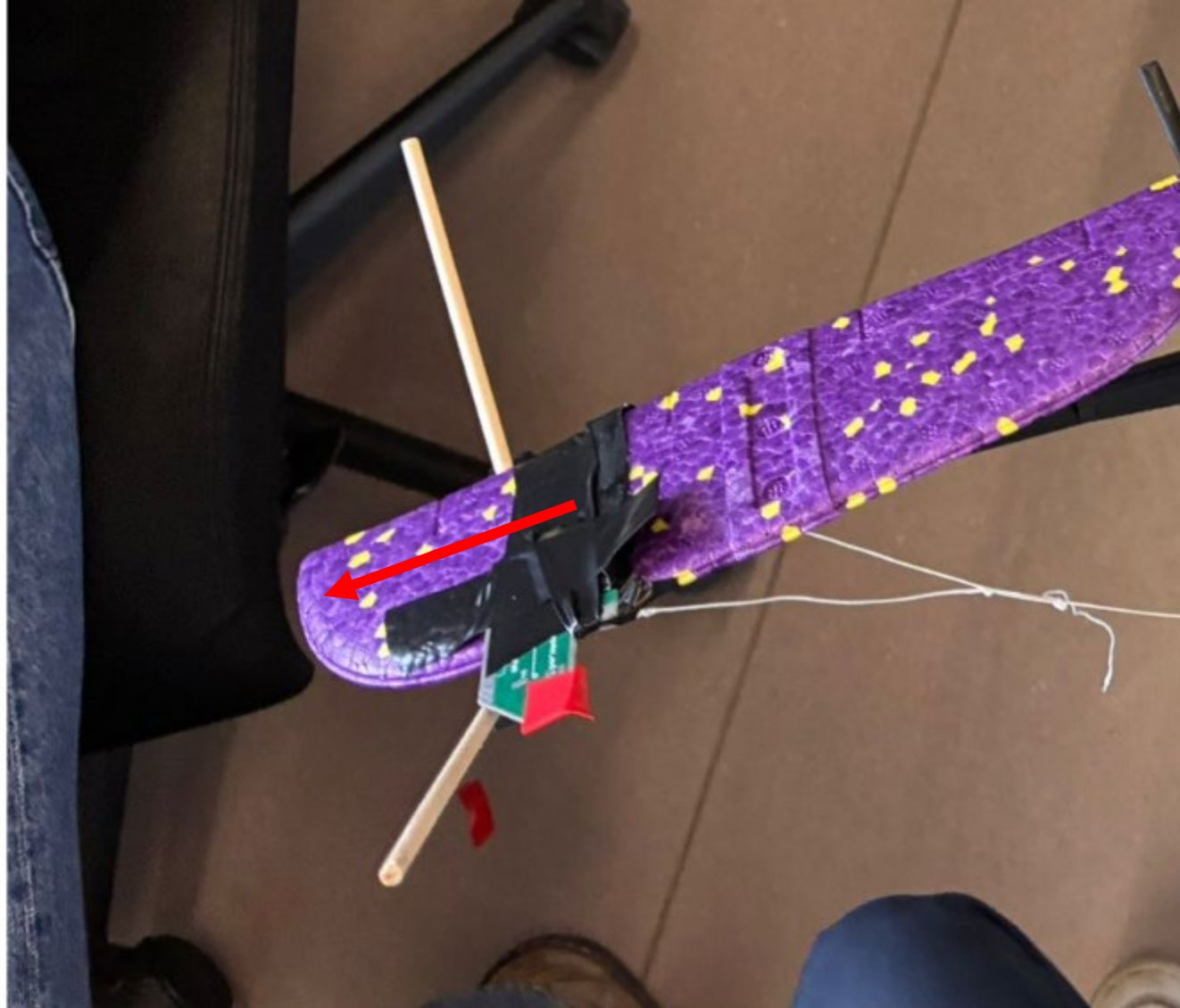
test → Odd behaviour



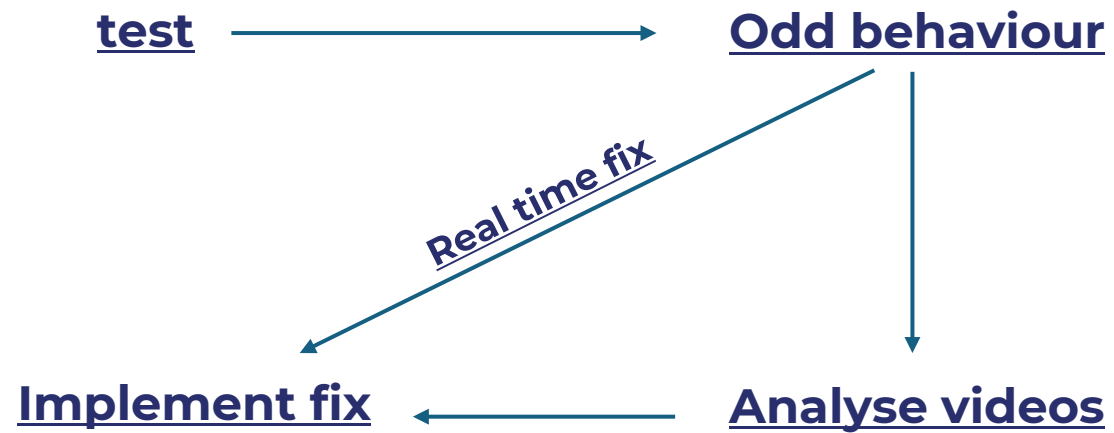
Testing



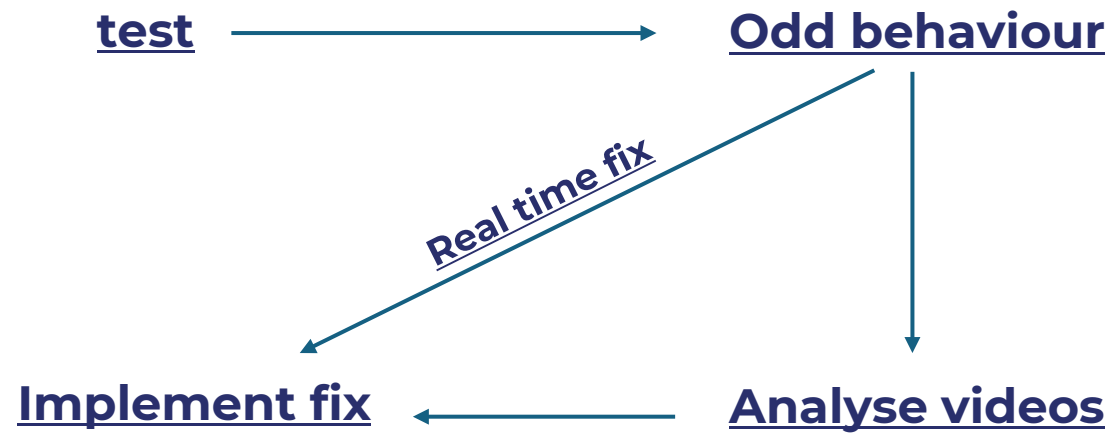
Testing



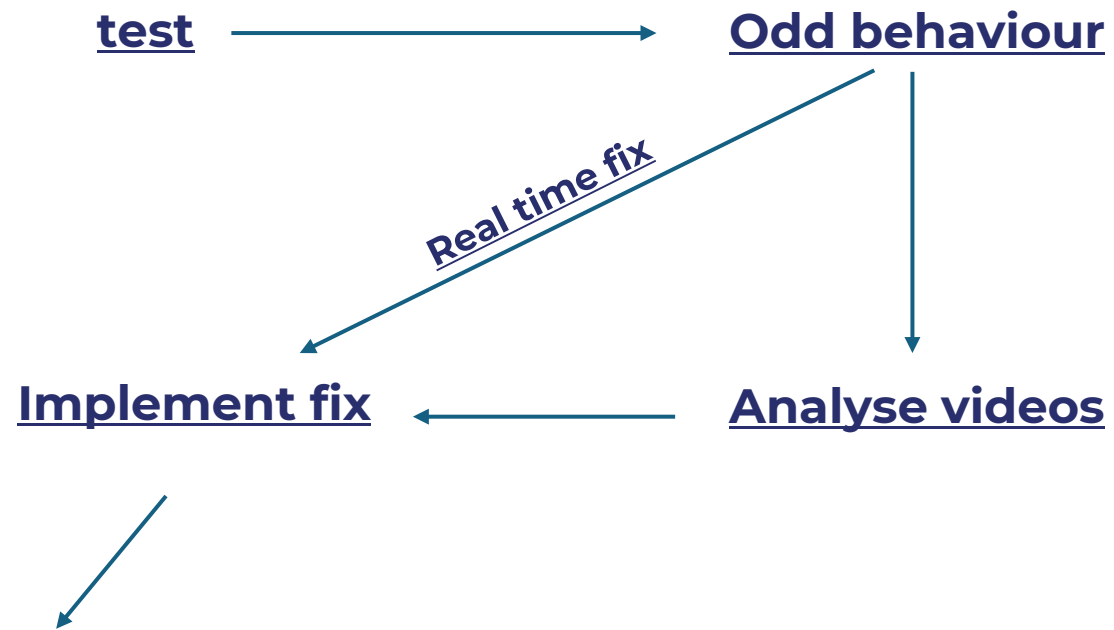
Testing



Testing

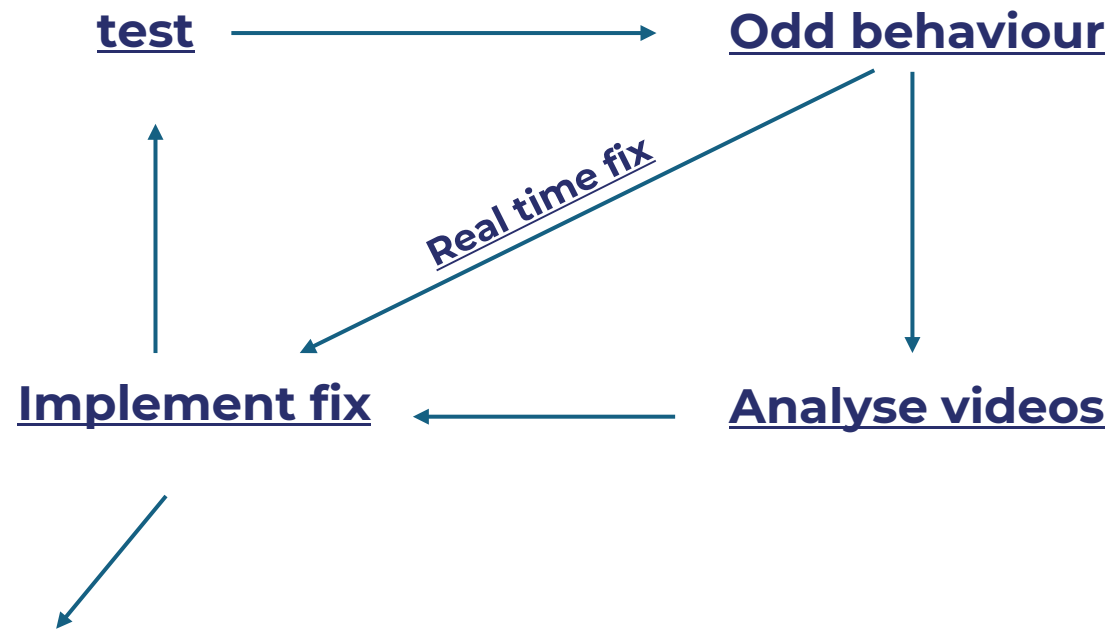


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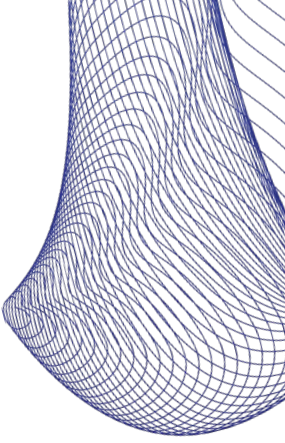
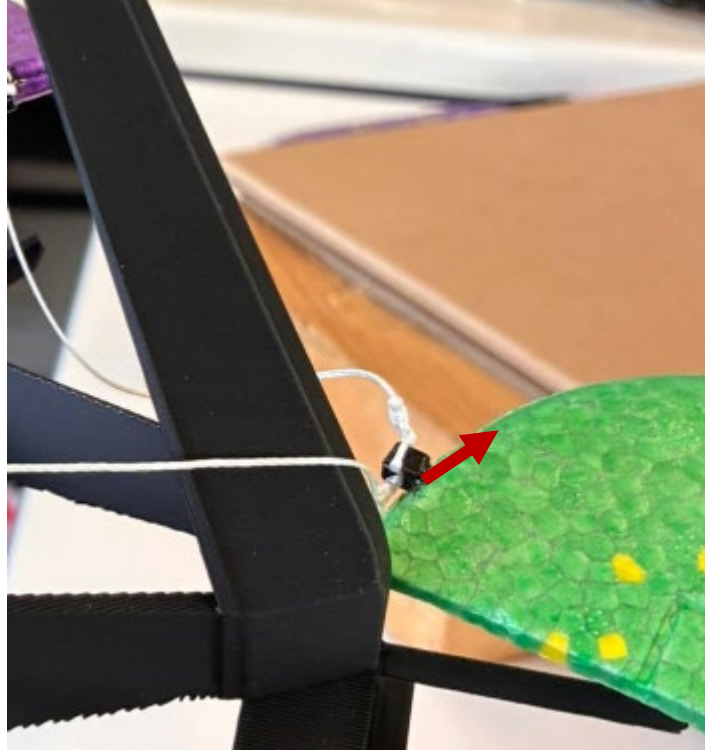
Usually tied to construction/integration defects/mismatches

Testing

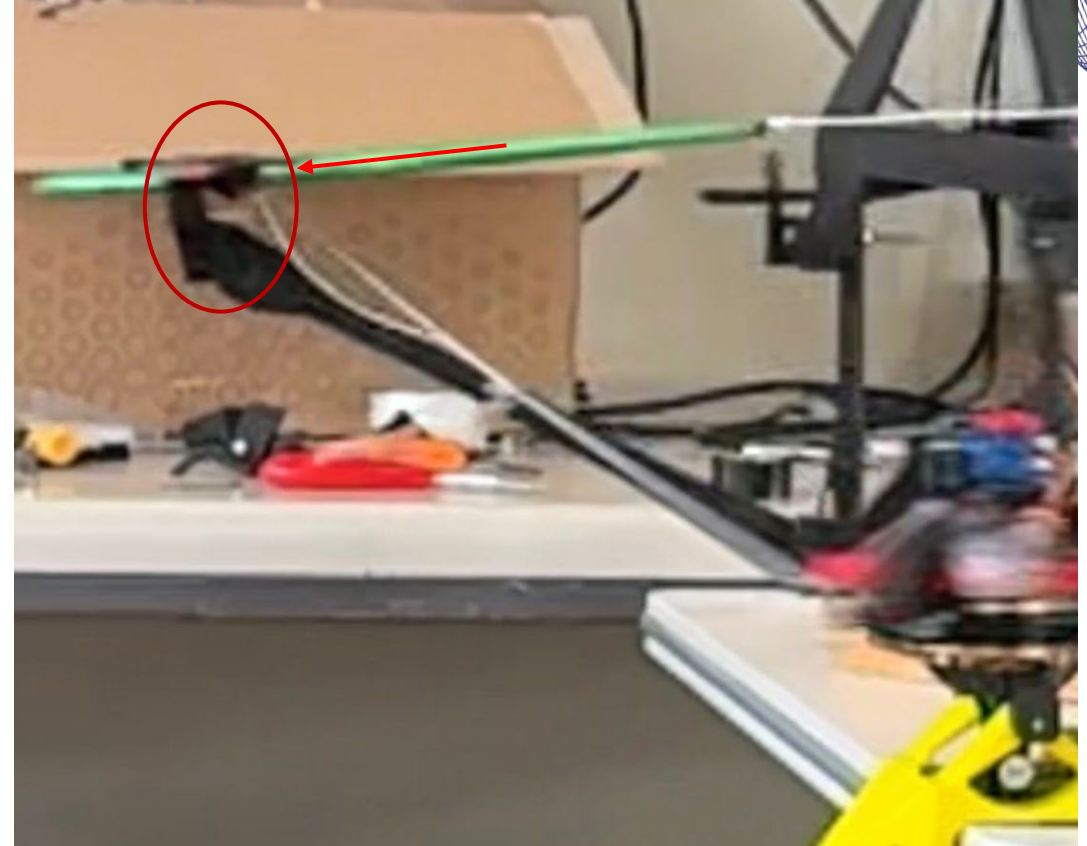


Usually tied to construction/integration defects/mismatches

Testing



Testing



Version 1: shortcomings

Vertical structural extension = added mass

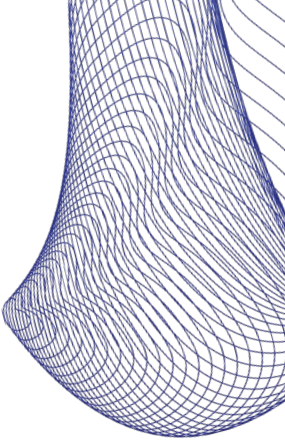
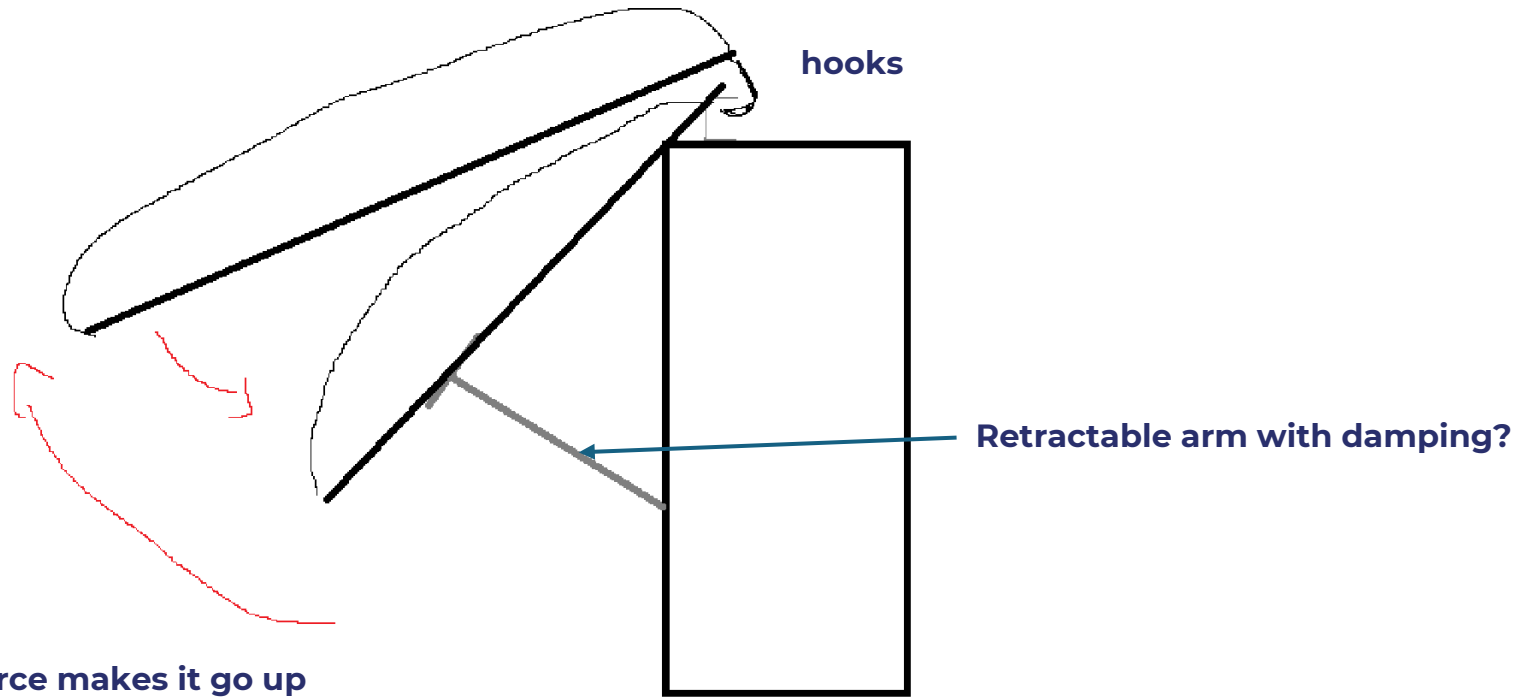


Radial structural extension = more inertia = higher hazard

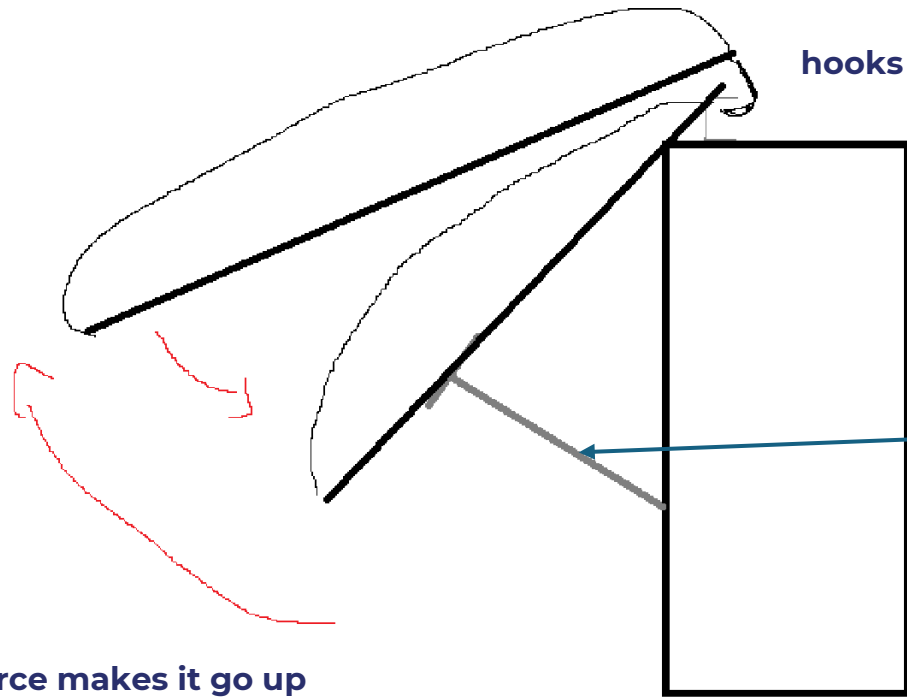
Future work

- Having a repeatable and clean TOL
- Automate the procedure
- Improve the current version

Future version?



Future version?



- Remove radial structure
- Reduce complexity
- Smaller surface exposed during storage

Centrifugal force makes it go up

Retractable arm with damping?

Completely vertical = higher structure $\propto$ wing length $\rightarrow$ angled

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Supervision: J. Waibel, Prof. C. Jones

THANK YOU FOR ATTENTION



<https://site.alphanemo.com/>